Original Article

Ketamine Therapy in Patients of Alcohol Use Disorder: A Prospective Pilot Study to Evaluate Abstinence Rate with This Add-on Treatment Modality

Nidhi Purushottum Marvania, Anto Leroy Rayan, Dhruti Vijaykumar Merchant, Drashti Paragkumar Shah, Prakash Ishwarlal Mehta, Fenil Dineshkumar Modh
Department of Psychiatry, GMERS Medical College and Hospital, Ahmedabad, Gujarat, India

Abstract

Background: Alcohol use disorders (AUDs) are emerging as a substance use disorder having various psychological and physical ill effects leading to hampering of individual's productivity and functioning. **Aims:** The aim of this study was to evaluate the efficacy of ketamine in enhancing confidence levels to abstain from alcohol. **Methods:** Eighteen patients were administered ketamine intravenously after detoxification; a total of three doses, with a 3-day gap between each. The Alcohol Abstinence Self-Efficacy Scale was used to measure their confidence in abstaining from alcohol. **Results:** The study revealed a significant increase in mean confidence scores after three ketamine sessions, with a final assessment showing a 51.98% increase in confidence from baseline. **Conclusions:** The study found that ketamine therapy significantly increased confidence levels in individuals with AUD to abstain from alcohol, with $P = 0.05$ threshold. However, the impact may wane over time, suggesting the need for additional support or maintenance treatments.

Keywords: Abstinence and craving, alcohol use disorder, ketamine therapy, paired *t*-test

Introduction

The Diagnostic and Statistical Manual of Mental Disorder-5 defines alcohol use disorders (AUDs) as a cluster of behavioral and physical symptoms, which can include withdrawal, tolerance, and craving. AUDs are among the most prevalent mental disorders worldwide, associated with psychiatric and physical comorbidities and contributing to global morbidity and mortality.[1]

The mechanism by which alcohol acts is by stimulating the inhibitory gamma-aminobutyric acid (GABA) receptor and inhibiting the excitatory N-methyl-D-aspartate (NMDA) receptor.[2] Craving for alcohol can be attributed as a result of an increase in the opioid particles and dopamine in the brain.[3] The synergistic effect of both these mechanisms is to produce sedation, euphoria, drowsiness, and physiological dependence to alcohol over time. There is an imbalance in inhibitory GABA and excitatory NMDA receptors on sudden abstinence of alcohol. A result of this imbalance is the manifestation of characteristic alcohol withdrawal syndrome (AWS) which consists of nausea, headache, tremors, tachycardia, hypertension, diaphoresis, hallucinations, and seizures.[4]

Food and Drug Administration approved anticraving drugs include naltrexone, acamprosate, and disulfiram.[5,6] Naltrexone acts as a mu-opioid receptor antagonist, whereas acamprosate reduces the cravings for alcohol by indirectly acting on GABA and NMDA receptors to reduce glutamate transmission.[6] Disulfiram interferes with alcohol metabolism by inhibiting the acetaldehyde dehydrogenase enzyme, thereby causing the aversive reaction to alcohol.[5] Although these empirical therapies have shown short-term effectiveness in reducing heavy drinking and increasing days of abstinence, they are still considered suboptimal given that only 25% of people achieve recovery in 1 year.[7] Given the scenario, there is a need to find additional efficacious pharmacotherapies to aid in the management of AUD and AWS.

Address for correspondence: Dr. Fenil Dineshkumar Modh, A-801, Dharti Saket Radiance, Nr. Vraj Residency 2, Bhadaj, Ahmedabad - 380 060, Gujarat, India.
E-mail: fenilmodh@gmail.com

Submitted: 06-Jun-2024 **Revised:** 07-Jun-2024 **Accepted:** 18-Jun-2024 **Published:** 01-Sep-2024

Access this article online

Quick Response Code:

Website: https://journals.lww.com/aips

DOI: 10.4103/aip.aip_110_24

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

How to cite this article: Marvania NP, Rayan AL, Merchant DV, Shah DP, Mehta PI, Modh FD. Ketamine therapy in patients of alcohol use disorder: A prospective pilot study to evaluate abstinence rate with this add-on treatment modality. Ann Indian Psychiatry 2024;8:324-7.

Ketamine is a potent, noncompetitive NMDA antagonist that has been widely accepted as a sedative and anesthetic, yet it shows various psychotherapeutic effects at subanesthetic doses.[8] Chronic alcohol exposure leads to upregulation of NMDA receptors which in turn is implicated in prefrontal circuit alteration, neurotoxicity, withdrawal, increased reactivity to substance cues, and cravings.[9] Possible hypotheses for the implication of ketamine in AUD are thought to be the normalization of cortical glutamate homeostasis and its action in inducing neuroplasticity, thereby facilitating learning and acquiring new skills.[10] Ketamine may provide a temporary boost to neurogenesis and synaptogenesis which may provide a window for psychological therapies to have better effectiveness.[11]

Methods

A prospective pilot study to evaluate the effectiveness of ketamine therapy in reducing alcohol craving among patients with AUD was initiated at GMERS Medical College and Hospital Sola, Ahmedabad, after approval from the Institutional Ethics Committee for Human Research-PG Research (IECHRPGR) with reference number GMERSMCH/IEC/98/2023. The sample was taken from those visiting the psychiatric outpatient department for de-addiction between November 05, 2023 and January 05, 2024, aged 18 years and above. Following regular detoxification procedures (injectable benzodiazepines and additional supportive care) targeting the acute withdrawal phase, patients were explained the ketamine therapy procedure and informed consent was obtained. A total of 18 patients were enrolled in the study after receiving general medical fitness for the procedure. Those with other substance use suffering from comorbid psychiatric disorders (except depression and anxiety spectrum disorders) and physically unfit were excluded from the study.

Patients received their first ketamine session on the 6th day after admission, followed by two subsequent doses at a 3-day interval. Ketamine was administered at a dose of 0.5 mg/kg diluted in 50 cc of normal saline intravenously over 50 min at a rate of 1 mL/min.

The study assessed alcohol use severity using Alcohol Use Disorders Identification Test (AUDIT) scores and Alcohol Abstinence Self-Efficacy Scale (AASE) to measure patients' confidence in abstaining from alcohol before and after ketamine sessions. AASE has 20 questions with 4 categories, namely negative affect; social/positive; physical and other concerns; and withdrawal and urges, with higher scores indicating greater confidence in abstaining.

Participants' baseline AASE scores were recorded before ketamine infusion, reassessed after each session, and discharged after the third session. A final assessment was conducted 2 weeks later. Data analysis involved calculating mean scores, comparing preketamine scores to postketamine scores, and analyzing changes in mean scores over time using paired *t*-test and ANOVA was used to determine the significance of changes in mean scores over time across the sessions for each category of the AASE scale.

Results

The study included 18 participants, 17 males and 1 female, with a mean age of 34.5 years. AUDIT score for all the participants was above 20. The statistical analysis, conducted through paired *t*-tests, demonstrated a significant increase in confidence levels among patients to abstain from alcohol following ketamine therapy. These findings are supported by $P = 0.05$ threshold across all comparisons [Table 1].

The result suggests the cumulative effect of ketamine therapy, where each additional session leads to further enhancements in confidence levels. The most remarkable improvement was noted after the third ketamine session. However, there was decrease in the final assessment AASE score, suggesting a diminishing effect over time without ongoing treatment.

The mean scores in each category of the AASE scale (negative affect, social/positive, physical and other concerns, and withdrawal and urges) showed a clear trend of increasing confidence levels from the preketamine session to the postketamine sessions [Table 2]. At the end of the study, seven participants had maintained abstinence from alcohol and the rest gave a history of consuming alcohol.

Discussion

This pilot study found that ketamine significantly enhances patients' confidence in abstaining from alcohol, potentially reducing craving in AUD patients. This serves as a testament to the studies carried out earlier which showed the impact of ketamine on AUD.[12-17]

In our study, we used the AASE score as an indicator of individual's craving for alcohol. AASE score helps to assess person's confidence in abstaining from alcohol during various circumstances by dividing the scoring into four categories:

Table 1: Paired *t*-test for preketamine versus postketamine session mean scores

Comparison	Mean preketamine session score	Mean postketamine session score	*T* statistics	*P*	Significance	Change (%)
Pre versus postsession 1	2.031	3.211	−6.985	<0.05	Significant	58.14
Pre versus postsession 2	2.031	3.792	−12.465	<0.05	Significant	86.73
Pre versus postsession 3	2.031	4.225	−22.037	<0.05	Significant	108.7
Pre versus final post 2 week	2.031	3.086	−4.982	<0.05	Significant	51.98

Table 2: ANOVA for four categories of Alcohol Abstinence Self-Efficacy Scale comparing means of preketamine versus postketamine sessions

Category	Preketamine session (mean)	Preketamine 1st session (mean)	Postketamine 2nd session (mean)	Postketamine 3rd session (mean)	Postketamine (2-week follow-up) (mean)	*T*-statistic (post 1st session)	*P* (post 1st session)	*T*-statistics (post 2nd session)	*P* (post 2nd session)	*T*-statistics (post 3rd session)	*P* (post 3rd session)	*T*-statistics (2-week follow-up)	*P* (post 2 weeks)
Negative affect	2.36	3.36	3.89	4.31	3.17	4.17	0.00064	7.43	<0.00001	12.93	<0.00001	3.48	0.00285
Social/positive	1.70	3.03	3.66	4.14	2.84	7.69	<0.00001	13.39	<0.00001	24.35	<0.00001	4.93	0.00013
Physical and other concerns	2.48	3.72	4.14	4.41	3.47	7.87	<0.00001	10.06	<0.00001	12.60	<0.00001	4.18	0.00063
Withdrawal and urges	1.59	2.73	3.48	4.03	2.87	5.19	<0.00001	10.65	<0.00001	17.06	<0.00001	5.39	0.00005

negative affect, social/positive, physical and other concerns, and craving and urges.[18] Participants' confidence increased with ketamine doses, but the follow-up AASE score decreased after 2 weeks, possibly due to diminishing ketamine effects. Despite this dip, the AASE score remained significantly positive compared to preinfusion baseline.

This increase in confidence of individuals of abstaining from alcohol may stem from ketamine minimizing the abstinence violation response.[17] This suggests the positive effect of ketamine on craving which stands with the findings of Yoon *et al*.[15] and Das *et al*.,[16] although it seems to be at contrast with Dakwar *et al*.[17] and Grabski *et al*.[12] where there was no significant difference noted among groups in craving and withdrawal.

During follow-up assessment, only 7/18 (38.8%) patients had maintained abstinence. Abstinence rates reported at the varied dosages of ketamine in different studies were:

- Krupitsky *et al*.: IM ketamine (3 mg/kg) + other interventions for an unspecified duration. Abstinence rate of 69.8% at 1-year follow-up[19]
- Krupitsky and Grinenko: IM ketamine (2.5 mg/kg) + other interventions for 3-month duration. Abstinence rate of 65.8% at 1-year follow-up[13]
- Kolp *et al*.: IM ketamine (2–3 mg/kg) + psychotherapy for 1–10-week duration. Abstinence rate of 70% for version/method 5 at 1-year follow-up[14]
- Dakwar *et al*.: IV ketamine (0.71 mg/kg) + motivational enhancement therapy (MET) for 5-week duration. Abstinence rate was reported at 53.9% at 21 days postinfusion.[17]

Our study's low abstinence rates, compared to previous studies, raise concerns about the validity of ketamine use for AUD, as reported abstinence rates of 28% can be achieved through MET or psychotherapy.[20] As mentioned by McIntyre *et al*., a possible hypothesis for ketamine action is by neurogenesis and neuroplasticity,[21] adjunctive treatment with psychotherapy may also help with more sustained and better results.[17] More data along with a longer study with adequate control; divisions for ketamine and MET would be required in a controlled setting to figure out if ketamine does have a positive benefit/harm ratio as compared to psychotherapy alone or not. A similar study had been done by Grabski *et al*., which has showed the beneficial effect of ketamine along with MET as compared to MET alone or ketamine alone.[12]

While ketamine has a durable impact, the effects seem to be diminishing over time as reported by decreasing AASE score. The dose of ketamine has been varied over all the similar studies done ranging from subanesthetic dosage of 0.35 mg/kg to 0.8 mg/kg IV or 2–3 mg/kg IM.[12-18] The range of ketamine dose used can produce varied results which may interfere with the interpretation of the results given by all studies. Given that our study has used a dose of 0.5 mg/kg, there is a scope to increase the dose of ketamine and evaluate if it produces improved results or not. There was no report of any adverse effects to ketamine in the dose of 0.5 mg/kg.

This pattern of our result highlights that each session of ketamine therapy contributed to a statistically significant improvement in confidence levels, with a clear cumulative effect observable across sessions. The final assessment suggests that while ketamine therapy has a durable impact, the effect may wane over time, highlighting the importance of considering additional support or maintenance treatments to prolong the benefits of ketamine therapy in the management of AUD.

Conclusions

This study is a critical research in the development of more effective treatment for AUD, one which allows patient in maintaining substance abstinence, thus improving the overall quality of life. Ketamine use can prove to be one such approach in addition medicine.

Statements

All data generated or analyzed during this study are included in this article. Further inquiries can be directed to the corresponding author.

Statement of ethics

- Study approval statement: This study protocol was reviewed and approved by the IECHRPGR with reference number GMERSMCH/IEC/98/2023
- Consent to participate statement: Written informed consent was obtained from participants (or their parent/legal guardian/next of kin) to participate in the study.

Data availability statement

All data generated or analyzed during this study are included in this article. Further inquiries can be directed to the corresponding author.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

References

1. Degenhardt L, Charlson F, Ferrari A, Santomauro D, Erskine H, Mantilla-Herrara A, *et al*. The global burden of disease attributable to alcohol and drug use in 195 countries and territories, 1990-2016: A systematic analysis for the global burden of disease study 2016. Lancet Psychiatry 2018;5:987-1012.
2. Schumann G, Saam C, Heinz A, Mann K, Treutlein J. The NMDA receptor system: Genetic risk factor for alcoholism. Nervenarzt 2005;76:1355-62.
3. Pastor R, Aragon CM. The role of opioid receptor subtypes in the development of behavioral sensitization to ethanol. Neuropsychopharmacology 2006;31:1489-99.
4. Jesse S, Bråthen G, Ferrara M, Keindl M, Ben-Menachem E, Tanasescu R, *et al.* Alcohol withdrawal syndrome: Mechanisms, manifestations, and management. Acta Neurol Scand 2017;135:4-16.
5. Jørgensen CH, Pedersen B, Tønnesen H. The efficacy of disulfiram for the treatment of alcohol use disorder. Alcohol Clin Exp Res 2011;35:1749-58.
6. Maisel NC, Blodgett JC, Wilbourne PL, Humphreys K, Finney JW. Meta-analysis of naltrexone and acamprosate for treating alcohol use disorders: When are these medications most helpful? Addiction 2013;108:275-93.
7. Fan AZ, Chou SP, Zhang H, Jung J, Grant BF. Prevalence and correlates of past-year recovery from DSM-5 alcohol use disorder: Results from national epidemiologic survey on alcohol and related conditions-III. Alcohol Clin Exp Res 2019;43:2406-20.
8. Gao M, Rejaei D, Liu H. Ketamine use in current clinical practice. Acta Pharmacol Sin 2016;37:865-72.
9. Nagy J. Alcohol related changes in regulation of NMDA receptor functions. Curr Neuropharmacol 2008;6:39-54.
10. Kalivas PW. The glutamate homeostasis hypothesis of addiction. Nat Rev Neurosci 2009;10:561-72.
11. Ivan Ezquerra-Romano I, Lawn W, Krupitsky E, Morgan CJ. Ketamine for the treatment of addiction: Evidence and potential mechanisms. Neuropharmacology 2018;142:72-82.
12. Grabski M, McAndrew A, Lawn W, Marsh B, Raymen L, Stevens T, *et al.* Adjunctive ketamine with relapse prevention-based psychological therapy in the treatment of alcohol use disorder. Am J Psychiatry 2022;179:152-62.
13. Krupitsky EM, Grinenko AY. Ketamine Psychedelic Therapy (KPT): A review of the results of ten years of research. J Psychoactive Drugs 1997;29:165-83.
14. Kolp E, Friedman HL, Young MS, Krupitsky EM. Ketamine enhanced psychotherapy: Preliminary clinical observations on its effectiveness in treating alcoholism. Humanistic Psychol 2006;34:399-422.
15. Yoon G, Petrakis IL, Krystal JH. Association of combined naltrexone and ketamine with depressive symptoms in a case series of patients with depression and alcohol use disorder. JAMA Psychiatry 2019;76:337-8.
16. Das RK, Gale G, Walsh K, Hennessy VE, Iskandar G, Mordecai LA, *et al.* Ketamine can reduce harmful drinking by pharmacologically rewriting drinking memories. Nat Commun 2019;10:5187.
17. Dakwar E, Levin F, Hart CL, Basaraba C, Choi J, Pavlicova M, *et al*. A single ketamine infusion combined with motivational enhancement therapy for alcohol use disorder: A randomized midazolam-controlled pilot trial. Am J Psychiatry 2020;177:125-33.
18. DiClemente CC, Carbonari JP, Montgomery RP, Hughes SO. The alcohol abstinence self-efficacy scale. J Stud Alcohol 1994;55:141-8.
19. Krupitsky EM, Grineko AY, Berkaliev TN, Paley AI, Tetrov UN, Mushkov KA, *et al*. The combination of psychedelic and aversive approaches in alcoholism treatment: The affective contra-attribution method. Alcohol Treat Q 1992;9:99-105.
20. Matching alcoholism treatments to client heterogeneity: Project MATCH three-year drinking outcomes. Alcohol Clin Exp Res 1998;22:1300-11. [doi: 10.1111/j.1530-0277.1998.tb03912.x].
21. McIntyre RS, Rosenblat JD, Nemeroff CB, Sanacora G, Murrough JW, Berk M, *et al*. Synthesizing the evidence for ketamine and esketamine in treatment-resistant depression: An international expert opinion on the available evidence and implementation. Am J Psychiatry 2021;178:383-99.